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# Human-machine collaboration for feature selection and integration to improve congestive Heart failure risk prediction

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ABSTRACT

The issue of harnessing machine learning (ML) algorithms for the prediction of adverse medical events is important considering the prevalence of vast repositories of patient-level medical data, and the quest to reduce healthcare costs while elevating the quality of care. Of particular interest is identifying a set of features that can better predict the risk of unplanned readmissions of chronically ill patients within 30 days post discharge, among them patients with Congestive Heart Failure (CHF). While numerous studies have employed ML algorithms to identify sets of significant features, only a handful have compared features extracted by various methods, and even fewer have compared the predictive power of feature sets selected by human experts to those selected by ML. Hence, this research aimed to elicit a comprehensive feature set selected by four ML algorithms and compare its predictive performance to that of a feature set selected by experts. We then evaluated a merged list of the two feature sets, constructing a human-machine collaborative set to predict the likelihood of 30-day unplanned readmission of CHF patients using data on 10,763 CHF patients hospitalized during the years 2010–2017. Our models achieved Area under the ROC Curve (AUC) above 0.8 and an Accuracy of ~0.89, comparable to the best models in the extant literature. Moreover, the Human-Machine Collaborative model significantly outperformed (best AUC 0.836) other human-only selected models as well as the machine-selected set. This study contributes to a better understanding of the power of using ML for patient risk stratification with special attention to the benefits of human-machine collaboration, even when these two entities work separately or in parallel, for improved clinical decision making.

## 1. Introduction

The widespread adoption of Health Information Technology (HIT) by healthcare organizations in many parts of the world has generated vast digital repositories of detailed patient-level data as a by-product of routine care delivery [105]. While the potential of these data to improve patient health outcomes and reduce healthcare costs is widely recognized [20,60,67,91,97,128,129], data accumulation, integration, access, analysis, visualization, and interpretation continue to be challenging, hindering the full realization of the potential of HIT and health data [49]. Providing high-quality care for complex, chronic conditions such as CHF is a high-dimensional, multi-level information processing problem that places significant demands on clinicians who are faced with the difficult task of efficiently identifying high-risk patients, deciding on appropriate interventions, and consistently monitoring an array of features which are associated with the disease and its progression [89,121]. There is a critical need to quickly retrieve, assimilate and assess laboratory test results, disease history, and other features both within and across patients [49,63,121]. Research has shown that individuals' ability to accurately integrate many disparate pieces of information for decision making is cognitively difficult [51]. When facing this information overload, people utilize a variety of simplifying heuristics which may bias or change judgments [3], likely resulting in suboptimal decision making [113].

Efficient and effective use of multi-dimensional, multi-sourced data for predictive analytics and risk stratification, particularly for clinical decision support at the point of care, is further exacerbated by the time-constrained and stressed environments where most of the acute care is provided – in the general practitioners' office, hospitals and emergency care institutions [49,82,111]. Predictive algorithms for risk assessment

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and point–of–care decision making have potentially an increasingly important role in supporting the day-to-day operations of healthcare organizations [3,76,84,99]. While generally true, this is particularly imperative for patients with complex chronic diseases such as Congestive Heart Failure (CHF), a common condition in patients with cardiovascular disease (CVD). Stratifying patients' risk level is an established practice in hospitals, where quite often cardiologists identify a set of indicative features for risk stratification, based on their expertise [9]. Other measures, described hereafter, use feature sets selected by machine learning (ML) algorithms, yet as far as we know, only a handful of studies have compared the predictive power of models based on the different feature sets, or examined the strength of human-machine merged sets in this context [9]. Therefore, the objective of this study is to address this lacuna by comparing the features selected by four ML algorithms with those selected by expert cardiologists, and then test the performance of predictive algorithms using these two sets, as well as a merged set. We show that the two feature sets are inherently different, and therefore merging them may have the advantage of reflecting the wisdom of both human experts and ML [101].

### 1.1. Leveraging medical data for treatment of congestive Heart Failure (CHF) patients

Heart Failure has been defined as a global pandemic, since it affects around 26 million people worldwide with annual costs exceeding $108 billion [29], primarily due to the high readmission and mortality rates of CHF patients [71]. Projections are even more alarming however, with total costs expected to increase by 127% by 2030 [104]. It is possible to highlight characteristics specific to high-risk patients using novel ML and other advanced statistical methods [18,19,21,22]. These insights may help physicians and hospital management identify high-risk patients at an early stage of the care process and apply appropriate interventions selectively [86,99]. Risk assessment of complex patients is a difficult, multi-level (population, individual and intervention levels of decision-making) and multi-dimensional (many variables under consideration at the same time) information processing task for clinicians facing a firehose of data and information [37,39,49,64,111]. Personalizing and communicating risk in dynamic, time-constrained, decision making environments such as day-to-day clinical care is even more challenging [89]. Carpenter, et al. [25] reviewed instruments used in EDs to assess risk of elderly patients and concluded that risk stratification of geriatric adults following ED care is limited by the lack of pragmatic, accurate, and reliable instruments, and that existing instruments designed to risk stratify older ED patients do not accurately distinguish high- or low-risk subsets. Yazdan-Ashoori, et al. [134] and Wang, et al. [127] tested the LACE (length of stay, acuity, comorbidities, and emergency department visits) index that predicts 30-day readmission or death (LACE$\geq$10 is considered high risk) for its predictive power. The former concluded that the threshold of LACE$\geq$13 is more suitable to identify high-risk patients, and that it has only a modest discrimination capability. The latter found that the LACE index may not accurately predict unplanned 30-day readmission for CHF patients. These and other studies highlight the difficulty encountered by clinicians where, in spite of theoretical availability of multitudinous data, only a fraction of these data can be used. Shilo, et al. [111] characterized various properties of medical data which they refer to as 'axes' and identified the potential and challenges of using big data in healthcare. They concluded that despite serious challenges, its potential is immense as patients' conditions and technologies become more complex over time. Hence, methods that create value for clinicians and patients, leveraging the risk prediction, are necessary for tackling the challenge of analytics consumption.

### 1.2. Readmission risk prediction and feature selection

30-day readmission, defined as an unplanned readmission to a hospital within thirty days post discharge, is an adverse event typical to chronically ill patients [38]. Frequent readmissions in a hospital are considered a signal of low quality treatment, therefore are penalized by the authorities in many countries in Europe and the US [48]. The problem of unplanned readmission of patients with chronic diseases is not new. An early readmission study [58] indicated that during 2003–2004 readmission rates were ~ 20% for 30-day readmission, and 34% for 90-day readmission. These rates remained fairly consistent during the following decade, with heart failure patients demonstrating the highest 30-day readmission rates, ~30% [61]. The cost associated with readmission is estimated around $26 billion annually [24]. Readmissions not only burden hospitals and elevate costs, but are also undesirable events for patients, who perceive it as a negative experience [69]. Researchers have harnessed advanced statistical and ML tools to develop accurate readmission prediction models, to advance more accurate risk stratification of hospitalized patients [29,59,112]. Traditionally, features for the models were selected by experts, for example the widely used LACE model [134], yet recently many prediction models used ML algorithms for feature selection [47,59]. Zhou et al. [140] identified 28 types of features used frequently in 73 reviewed models, the most prevalent being comorbidities (in 54 models, 74%), demographic/social features (in 45 models, 62%), length of stay, and number of previous readmissions (each one in 29 models, 40%). Other used features were laboratory tests (in 25 models, 34%), and medications (in 21 models, 29%). By and large, features can be roughly grouped into eight categories: patient demographics and socioeconomic status including type of insurance, procedures, diagnoses, medications, lab tests, clinical measures, visit history, and hospital characteristics [59]. Table 1 summarizes the distribution of features in thirteen exemplary publications, displaying similar distributions as in Zhou et al. [140]. The three most frequently used feature categories are diagnoses (or comorbidities), socio-demographic, and visit history. Lab results were used in <40% of the reviewed studies, yet, with the prevalence of integrated health IS in many hospitals [1,53], lab results should be more accessible for feature selection, potentially reflecting the patients' health status on a continuum rather than as a dichotomous variable [137].

Two studies examined the performance of readmission prediction models via the various feature categories. Shameer, et al. [109] compared models predicting 30-day readmission based on a single-category features set. They found that the lab results category performed worse than the other models. In contrast, Jiang, et al. [59] found that the dynamic clinical predictors capturing laboratory results and vital signs had the highest predictive value compared with demographic, administrative, medical, and procedural data.

While feature selection methods varied much among the studies, the AUC converged around 0.7, usually between 0.6 and 0.75. Zhou et al. [140] found only two models with AUC above 0.8 [35,110].

Overall, numerous studies have harnessed ML methods to find sets of features affecting the likelihood of hospital readmission of CHF patients and patients with other chronic diseases [47] using EHR-based data. Fewer studies, however, have compared the performance of models based on human selected features to those based on ML selected features or to the performance of predictive models that include both human and ML selected features. By choosing human-selected features only, a vast amount of information captured and stored in the EHR is lost to the models, whereas relying only on ML features ignores human intelligence and knowledge [9].

### 1.3. Human-machine collaboration for ML-based prediction

Collaboration is defined as the process through which two or more actors engage in a constructive management of differences in order to define common problems and develop joint solutions [50]. The term 'human-machine collaboration' is often used in the literature to describe synchronous work of humans and sophisticated machines, where individuals complete tasks requiring intuition and expertise lacked by the

**Table 1**
Feature category distribution and AUC.

| Publication | Feature selection method | Num. of Features | Demographic | Diagnoses | Procedures | Medications | Lab tests | Vitals | Visit history | Other | AUC |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Amarasingham et al. [7] | 4-stage regression model | 12 | 5 | 1 | 1 | | | | 2 | 3 | 0.72 |
| Ashfaq et al. [9] | Experts | 11 | 2 | 1 | 1 | 1 | | | 6 | | 0.77 |
| Bardhan, et al. [10] | BG/EG Hurdle model | 23 | 4 | 9 | 1 | | | | 2 | 7 | 0.601 |
| Basu Roy, et al. [12] | Random forests | 10 a* | | 6 | | | | 1 | | 3 | 0.625 |
| Bayati, et al. [14] | LASSO | 14 b* | | 3 | | | 8 | | 2 | 1 | 0.69 |
| Ben-Chetrit, et al. [23] | LR | 9 | 2 | 2 | | | 4 | | | 1 | 0.7 |
| Golas, et al. [44] | Deep Unified Networks (DUNs) | 15 b* | | 11 | | | 3 | | 1 | | 0.764 |
| Graham, et al. [45] | LR | 22 | 8 | 6 | 6 | | | | 1 | 1 | N/A |
| Rico, et al. [100] | LR | 7 | 1 | 1 | | | | | 1 | 3 | 0.63 |
| Rico, et al. [96] | Cox PH | 9 | 3 | 1 | | | | | 2 | 3 | N/A |
| Shameer et al. [105] | Naïve Bayes | c* | | d* | 4 | 3 | 4 | | | | 0.78 |
| Turgeman and May [122] | Ensemble | 18 | | 5 | | | 2 | | 10 | 1 | N/A |
| Times used (%) | | | 8 (61.5) | 13 (100) | 5 (38.5) | 3 (23) | 5 (38.5) | 1 (7.7) | 10 (77) | 9 (69) | |

Notes: a*: 10 most affective features of 16. b*: most important features. c*: 105 features were used, number of most affective features not available. d*: several features from this category (not detailed). N/A – Not available.

machine [139]. The human–machine collaboration can be optimal when actors complement each other's strengths [62,130,132]. For example, while machines only apply logic to solutions and decisions, humans combine intuition, expertise and nuances in their solutions, which may balance biases caused by the algorithms [81,101]. Human–machine collaboration has three perspectives: 1) the organizational perspective, 2) the relationship perspective, and 3) the interaction perspective [132]. The organizational perspective relates to the smooth assimilation of the systems in the routine business processes. The relationship perspective deals with emotional and cognitive aspects of people interacting with machines, and the third, interaction perspective looks into optimal modes of interaction between the two entities [132] to achieve compliance and synergetic human-machine collaboration [62]. In the medical domain, Mumtaz, et al. [88] described such a collaboration in a futuristic operating room, where artificial intelligence (AI) algorithms assist surgeons in tedious, data intensive operations, thereby alleviating cognitive burden. As robotic surgery becomes more prevalent and thoroughly evaluated in situ, it provides a practical demonstration of human-machine collaboration, with its own successes and challenges [46]. Firms are using ML-based decision support systems in several businesses aimed at enhancing the quality of decisions and their overall performance [52]. However, quite often, the support and intervention of human agents are needed to attain superior performance [43,101,130]. Ashfaq, et al. [9] found that a model composed of combined human and machine selected features outperformed each of the individual models. The improvement was achieved by adding only 10 human-selected features that included mostly demography and utilization features, and only one clinical feature.

In the current study, however, we describe a non-interactive concept of human-machine collaboration, where humans and machines select feature sets separately, utilizing their unique types of expertise. It then makes sense, since the results are inherently different, to merge them and compare the performance of the new 'collaborative' set, against the performance of the two separate feature sets.

### 1.4. A two-stage process for developing prediction models

Predicting the likelihood of an event's occurrence using ML algorithms commonly involves running the algorithms on a large segment of the dataset for training and optimization, and then executing it on the remaining smaller segment of the dataset to test the performance of the model [135]. The shortcoming of this method is that some of the ML algorithms do not identify the feature list and the importance of the various features. Thus, using such algorithms hinders the ability to glean wisdom from the results, beyond the predictive power. Moreover, massive multi-dimensional datasets require strong computing power and might take a long time to completion [27]. Therefore, a two-stage analysis is frequently applied which first reduces the dimensionality of the dataset using robust ML algorithms for feature selection, and then executes the prediction models on a reduced set of features. This two-stage process simplifies the development of the prediction model and enables examination and comparison among various feature sets. This analysis is common in many areas beyond the medical field, such as in weather forecasting [107], stock performance [136], prediction of defect-prone software [6], financial statement fraud detection [98], bankruptcy prediction [68], and Netflix movie ratings [138]. In the medical field, where identification of features affecting adverse events are of great interest, the first-stage FS strategy is used quite often. Several examples can be found in Table 1 that summarizes the feature distribution of thirteen studies. Thus, the current study implements a process that is widely accepted in predictive analytics in the medical field and beyond.

## 2. Data and methods

This section discusses the research flow from raw data to results, highlighting the methods employed in the various steps. Starting with the research dataset from the comprehensive Sheba data warehouse with many variables, we performed dimensionality reduction using both human experts and feature selection methods to create three feature sets: expert list, ML list and a merged list. The three feature sets were compared on prediction power using two classification algorithms.

### 2.1. The study setting

The Sheba Medical Center is the largest Israeli hospital, located in the center of the country and ranked among the top hospitals worldwide [85]. It is a leading public academic medical center, annually handling over a 1.5 million patient visits and ~ 200 K emergency visits, and conducts more than two million medical tests of all types. Sheba is an extensively computerized hospital, where an integrated EHR system is used in the Emergency Department (ED) and all inpatient departments, fully replacing paper-based medical records. Each patient admitted to the hospital, via the ED or otherwise, who is diagnosed with a cardiovascular condition as a primary diagnosis, has their data uploaded to the data warehouse. The records have the potential to contain above

thousand features for each patient, albeit sparsely populated, with many data challenges. Such a health data repository at the patient level is both rich and unique for large-scale data analysis, but challenging to utilize at the point of decision making.

In this study, we use the Sheba data warehouse to extract the full hospitalization details pertaining to patients treated at Sheba from 2010 to 2017 with CHF as their primary diagnosis.

### 2.2. Step 1: data engineering and dimensionality reduction

The Sheba Heart Center data warehouse stores the records of all patients hospitalized with a cardiovascular disease as a primary diagnosis as of 2010. A set of Python/Pandas/T-SQL programs was used to read data from the server and process them to a representation that is most suitable for the current study. First, we included only patients who were diagnosed with CHF as their primary diagnosis (ICD-9 code 428). Second, diagnoses were automatically detected and encoded using either the text description (which vary from system to system) or the ICD-9 encoding, where possible. Patients' chronology of hospitalizations and visits were indexed by an anonymized patient ID so that potential outcomes (e.g. readmissions) can be evaluated.

Although all the data sources belong to the same institution, since the systems are disparate, heavy data manipulation was still required to achieve a unified data representation across all systems. The initial database consisted of ~1200 features requiring intensive feature engineering. A pre-feature selection process was performed, excluding sparsely populated variables with substantial missing values, textual variables with notes that should be parsed, as well as categorical (Y/N) variables of numerous medication and supplements (e.g. vitamins) unrelated to CHF. Variable removal was thus based on missing values and clinical judgment. Imputation was carried out using the Predictive Mean Matching method [72] with the 'mice' package in R [124]. The resulting dataset included 10,763 patients, with only the first hospitalization included for each patient. The dependent variable was '30-day readmission' and was assigned the value '1' if the patient was readmitted to the hospital within 30 days post discharge for the same primary diagnosis (CHF), after the first hospitalization. The dataset included 134 features after the first cleaning.

### 2.3. Step 2: correlation tests

We ran three correlations tests: Pearson, Spearman, and Kendall's Tau for the various types of variables. The results of the Spearman test revealed 11 groups of correlated features, covering those found by the other two tests as well. Consequently, 14 features were discarded for high correlation ($r \geq 0.7$) with other features, based on domain knowledge.

### 2.4. Step 3: feature selection (FS)

Feature selection by experts is a common practice, widely covered in the literature [126].

Human experts: Two expert cardiologists received the 120 features retained in the dataset and were asked to select the most critical features to assess 30-day readmission risk of CHF patients. They were not limited to any number of features. The selection process was conducted without seeing the ML feature set, by consensus between the two experts, based on their clinical expertise and the extant literature. The two experts served as the Head of the Cardiology Department and the Head of the Heart Rehabilitation Center at the Sheba Medical Center. They selected the feature set together. We termed this feature set "the expert set".

ML algorithms: four ML algorithms were employed for FS: logistic regression -ridge, logistic regression – lasso, random forest – Gini index and random forest – information gain.

#### 2.4.1. Logistic regression-ridge (L2 norm)

This method is useful when there are many covariates, and when covariates might be correlated. The ridge parameter penalizes coefficients of covariates that cause high variability in the model by setting a constraint on the sum of squared coefficients, setting them close to zero, thereby actually leaving them in the equation but minimizing their effect [79].

#### 2.4.2. Logistic regression-Lasso (L1 norm)

In contrast to ridge, Lasso [83,90,119] shrinks some coefficients to zero by setting a constraint on the sum of the absolute values of the coefficients of standardized covariates. In the specific case of readmission, logistic regression-lasso explores the linear hyperplane as the decision boundary for patients with high risk versus low risk of readmission [114].

#### 2.4.3. Random forest-Gini Index

Random forest allows looking at feature importance, which is indicated by how much the Gini Index for a feature decreases at each split. The more the Gini Index decreases for a feature, the more important it is. We used random forest with Gini Index as the impurity splitting technique as in Liu, et al. [77].

#### 2.4.4. Random forest-information Gain (IG)

Differently stated, IG measures the amount of information in bits about the class prediction, if the only information available is the presence of a feature and the corresponding class distribution [102].

We used randomized search along with cross validation for hyperparameter tuning for all four methods.

#### 2.4.5. Previous studies using multistage analysis

XGBoost was used for prediction using the three previously selected feature sets: 1) feature set selected by experts, 2) feature set selected by four algorithms, and 3) the merged list of the two feature sets

Such a two-stage process of selecting features first and then testing the predictive power of the set using another algorithm, has been applied in prior studies (see section 1.4).

### 2.5. Step 4: prediction modelling

XGBoost (Extreme Gradient Boosting), originally created from the concept of gradient boosting presented by Friedman [42], was further developed by Chen and Guestrin [28]. The basic principle of gradient boosting is a straightforward method to build a new model in a gradient direction of the residual errors in order to minimize the loss function which is generated at each iteration [74]. XGBoost trains a series of weak tree models, of which the latter learner focuses more on the errors of the former learner. It performs a second-order Taylor expansion on the loss function, and it can automatically use the multithreading of the CPU for parallel computing. Besides, XGBoost uses a variety of methods to avoid overfitting. We used randomized search with cross validation for hyperparameter tuning. We tuned the XGBoost model on 10 hyperparameters, including: min child weight, max depth, number of estimators, gamma, subsample, colsample by tree (tree-specific parameters), scale positive weight (number of negative cases to the number of the positive cases), lambda, alpha and eta (the regularization parameters). Then, we trained the XGBoost model with the optimal hyperparameter setting with the resampled training dataset.

## 3. Results

In this section, we describe the data analyses including five steps: 1) multivariate correlation tests, 2) feature selection, 3) prediction modelling, 4) model performance analysis, and 5) robustness checks using AdaBoost algorithm and Neural Network as classifiers (see Online Appendix A).

10,763 patients hospitalizations were included comprising the first occurrence for each patient. Their records included a sufficient number of features for this study and could be reliably tracked for readmissions. Of the 10,763 patients, 1026 (9.53%) were readmitted within 30 days post discharge. The dataset included 134 features with 88 numerical features and 46 categorical features before correlation tests.

### 3.1. Correlation tests

We calculated covariate correlation coefficients using the Pearson, Spearman, and Kendall's Tau tests. We defined a pair of values as highly correlated if its absolute correlation value was higher or equal to 0.7, retaining only one variable from a highly correlated set based on domain knowledge. This step resulted in discarding 14 variables, leaving a total of 120 features for subsequent analysis.

### 3.2. Feature selection

The set of 120 variables included features from the following domains: 1) demographic/personal, 2) medication, 3) lab results, 4) ECHO and 5) comorbidities or past diagnoses.

#### 3.2.1. The expert list

The expert list of 26 features included age, gender, differential diagnoses, current and past health indicators, medications, lab and imaging results. Table 2 Summarizes the descriptive statistics of the features for the two cohorts: Those who were readmitted within 30 days post discharge (event = 1, 1026 patients, 9.53%) and those who were not (event = 0; 9737 patients, 90.47%). All but five features significantly differed between the two cohorts, marked with asterisks in Table 2. The five insignificant features were age, BMI, data blocker agent treatment, lactate, and SGOT/AST level in the blood

Of the 26 features selected by the experts, three were personal details, three were medication related, eight described comorbidities, three were heart functioning indicators and nine were lab results (Table 3). It is thus evident that the expert list covered several data domains, but mainly focussed on comorbidities and lab results (17 altogether).

#### 3.2.2. Feature selection – machine learning (ML) list

before building the predictive model, we reduced the dimensions of the dataset using four feature selection methods: Logistic regression – Ridge, logistic regression – Lasso, random forest using the Gini index and using information gain. Before processing the prediction analytics, we scaled the dataset using MinMaxScaler method in the sklearn package [92] to help convergence and better interpretation. For each of the methods, we sorted all the 120 features by importance as ranked by the algorithm, from 1 to 120. Features ranked lower than 50 were discarded. We then included in the ML feature list only features ranked as top 50 by all four methods, resulting in 21 features (Table 3)

As seen in Table 3, the rankings of the various features differed among the four methods, yet the domains from which the features were selected were similar: ECHO and lab results. As opposed to the experts list, neither medications, nor comorbidities or personal traits were included. Moreover, the intersection of the lab results in both lists includes only SPAP, LDH and Potassium. These attest to the significant differences between the experts and the ML lists, providing additional motivation for examining the performance of a merged list.

#### 3.2.3. The merged list

We joined the ML and the expert lists, forming the human-machine collaborative list that included 42 features, 5 of which were selected by both ML and experts (ECHO: Mitral valve deceleration time, ECHO: Mitral valve peak e-wave, lactate dehydrogenase (LDH) level in the blood, potassium level in the blood, and SPAP). While the experts selected both lab tests and diagnoses, the ML mainly selected lab tests, thus joining the two lists balances this bias. Notably, the collaborative list is more balanced than the ML list in terms of the feature domains (Table 4)

**Table 2**
Descriptive statistics of the expert list.

| Number of patients included ($N$ = 10,763) | | | 9737 (90.47%) | 1026 (9.53%) | |
|---|---|---|---|---|---|
| Type | No. | Indicator | Not readmitted (0) | Readmitted (1) | $P$-Value |
| | 1 | Gender (female) (%)* | 46.42% | 43.2% | 0.016 |
| | 2 | Age | 76.99 ± 12.66 | 76.96 ± 11.65 | 0.465 |
| Personal | 3 | Body Mass Index (BMI) | 30.56 ± 87.7 | 30.49 ± 41.8 | 0.487 |
| | 4 | ACE inhibitor treatment (Y)*** | 56.46% ± 49.58% | 63.56% ± 48.14% | <0.001 |
| | 5 | Aldacton treatment (Y)*** | 37.16% ± 48.14% | 53.2% ± 49.91% | <0.001 |
| Medi-cation | 6 | Beta Blocker Agents treatment (Y) | 31.48% ± 46.45% | 33.64% ± 47.26% | 0.085 |
| | 7 | History of Percutaneous Coronary Intervention (PCI)*** (Y) | 17.56% ± 38.05% | 25.53% ± 43.62% | <0.001 |
| | 8 | Past history of Myocardial Infarction (MI)*** (Y) | 27.76% ± 44.78% | 43.02% ± 49.53% | <0.001 |
| | 9 | Past history of Atrial Fibrillation*** (Y) | 45.57% ± 49.81% | 55.39% ± 49.72% | <0.001 |
| | 10 | Past history of Hypertension*** (Y) | 70.55% ± 45.58% | 76.66% ± 42.31% | <0.001 |
| | 11 | Has Diabetes Mellitus (DM)*** (Y) | 41.38% ± 49.25% | 49.6% ± 50.01% | <0.001 |
| | 12 | Anaemia*** (Y) | 37.44% ± 48.4% | 54.42% ± 49.82% | <0.001 |
| | 13 | Has congestive obstructive pulmonary disease(COPD)***(Y) | 50.68% ± 50% | 68.13% ± 46.12% | <0.001 |
| Comorbidities | 14 | Has renal disease *** (Y) | 45.64% ± 49.82% | 68.86% ± 46.32% | <0.001 |
| | 15 | Heart ECHO test: Mitral Valve Peak e-Wave*** | 81.82 ± 37.88 | 66.55 ± 40.69 | <0.001 |
| | 16 | Heart ECHO test: Mitral Valve deceleration time*** | 188.7 ± 61.04 | 166.4 ± 56.98 | <0.001 |
| Heart Condition | 17 | Ejection Fraction*** | 48.18 ± 20.34 | 43.92 ± 19.9 | <0.001 |
| | 18 | Sodium level in the blood*** | 138.77 ± 5.65 | 137.63 ± 5.08 | <0.001 |
| | 19 | Systolic Pulmonary Artery Pressure (SPAP)*** | 47. 1 ± 10.16 | 49.25 ± 10.71 | <0.001 |
| | 20 | Potassium level in the Blood*** | 4.38 ± 0.71 | 4.6 ± 0.74 | <0.001 |
| | 21 | Lactate Dehydrogenase (LDH) level in the blood | 20.86 ± 9.94 | 20.49 ± 9.94 | 0.175 |
| | 22 | Albumin level in the blood*** | 3.44 ± 0.55 | 3.63 ± 0.45 | <0.001 |
| | 23 | Haemoglobin level* | 11.54 ± 1.94 | 11.67 ± 1.93 | 0.011 |
| | 24 | SGOT/AST in the blood (units/L) - liver disease measure | 31.37 ± 23.80 | 32.05 ± 23.56 | 0.287 |
| | 25 | C-Reactive Protein (CRP) level in the blood*** | 44.76 ± 46.58 | 40.69 ± 45.29 | <0.001 |
| Lab Results | 26 | GFR/MDRD- kidney disease measure*** | 52.13 ± 28.41 | 46.67 ± 22.88 | <0.001 |

Notes: 1) Statistical significance is indicated by:'*' for $p < 0.05$;'**' for $p < 0.01$; and'***' indicating $p < 0.001$. 2) Data are the mean (±SD) or number of subjects (proportion).

**Table 3**
Features selected by the four ML algorithms and their ranking.

| Domain | No. | Description | LR -Ridge | LR -Lasso | RF- Gini Index | RF - IG |
|---|---|---|---|---|---|---|
| Heart Condition | 1 | ECHO: Estimated RAP | 3 | 3 | 1 | 1 |
| ECHO | 2 | ECHO: Interventricular septal end thickness | 26 | 25 | 19 | 18 |
| | 3 | ECHO: Left ventricle fractional shortening | 46 | 34 | 10 | 17 |
| | 4 | ECHO: Mitral valve deceleration time | 11 | 10 | 5 | 3 |
| | 5 | ECHO: Mitral valve peak a-wave time | 21 | 17 | 2 | 2 |
| | 6 | ECHO: Left ventricular diastolic dimension | 7 | 7 | 9 | 9 |
| | 7 | ECHO: T1 gradient | 18 | 18 | 8 | 12 |
| | 8 | ECHO: Mitral Valve peak e-wave | 44 | 33 | 6 | 4 |
| Lab Results | 9 | Lactate Dehydrogenase (LDH) level in the blood | 28 | 24 | 22 | 20 |
| | 10 | Troponin level in the blood | 13 | 13 | 31 | 25 |
| | 11 | Calculated Osmolality | 1 | 1 | 43 | 19 |
| | 12 | Partial pressure of CO2 (PCO2) in the blood | 9 | 15 | 27 | 30 |
| | 13 | Glucose level in the blood | 2 | 2 | 16 | 16 |
| | 14 | Mean Platelet Volume (MPV) in the blood | 39 | 44 | 15 | 23 |
| | 15 | Total Cholesterol | 25 | 19 | 40 | 50 |
| | 16 | Systolic Pulmonary Artery Pressure (SPAP) | 14 | 11 | 39 | 22 |
| | 17 | Partial Pressure of Oxygen (PO2) in the blood | 29 | 23 | 45 | 34 |
| | 18 | Potassium level in the blood | 10 | 14 | 48 | 37 |
| | 19 | Serum Glutamic Pyruvic Transaminase (SGPT)/ Alanine Aminotransferase (ALT) | 23 | 20 | 37 | 39 |
| | 20 | Ionized Calcium level in the blood | 36 | 32 | 24 | 45 |
| | 21 | Base Excess level of the blood | 40 | 40 | 46 | 43 |

**Table 4**
Distribution of the feature sets (number of variables) among the six categories.

| | Expert List | ML List | Human-Machine Collaborative |
|---|---|---|---|
| Personal/Demographic | 3 | 0 | 3 |
| Medication | 3 | 0 | 3 |
| Labs | 9 | 13 | 19 |
| Echo | 3 | 8 | 9 |
| Comorbidities | 8 | 0 | 8 |
| Total | 26 | 21 | 42 |

### 3.3. Prediction model

The best XGBoost model was the one tuned for the best ROC-AUC. We also conducted the DeLong [33] test for the three lists. The target variable was readmission within 30 days. We split the dataset into a training dataset and a hold-out sample for testing using an 80/20 split, with the training dataset randomly shuffled for five-fold cross validation. This training dataset includes 8610 patient records, while the hold-out sample, or testing dataset, included 2153 patient records. Visualizing the distribution of the outcome variable, 30-day Readmission, in the training dataset showed that the dataset was imbalanced, with 7787 negative (not readmitted) cases and 823 (9.56%) positive (readmitted) cases (Fig. 1). The impact of dataset imbalance on prediction modelling was adjusted using SMOTE to resample the data. After resampling, the new training dataset included the same number of negative and positive cases of 7787.

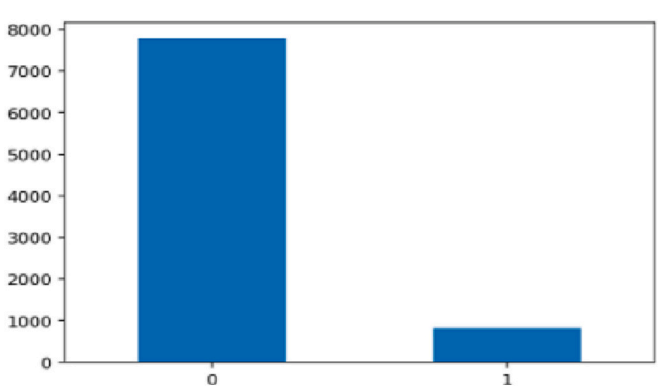


**Fig. 1.** Data before resampling (0-no readmission; 1- readmission).

### 3.4. Model performance analysis

The XGBoost algorithm was executed for Readmission for each of the three feature lists, with scaled features. Table 5 summarizes the results of predicting the occurrence of 30-day readmission. All the lists performed quite well, achieving high AUC compared to the results in the literature. As evident in Table 5 and in Fig. 2, the ML list outperforms the expert-list in terms of AUC (AUC = 0.8156 vs. 0.7116 respectively), whereas the human-machine collaborative list slightly improves the performance of the ML list, yet significantly outperforms the experts list ($p$-value<0.001, chi = 36.95, AUC = 0.8289) [33]. The human-machine collaborative list achieved the best AUC of 0.8289, showing 1.6%–16% improvement in the AUC values over the other sets. Fig. 2 shows the ROC curves and AUC of the three feature lists.

To ensure completeness, we have included the results of the prediction model using all 120 variables in Table 5. As expected, incorporating additional features marginally enhances the model performance in terms of AUC and accuracy, with improvements of 1.07% (from 0.8289 to 0.8378) and 0.15% (from 0.8876 to 0.8889), respectively. However, since the main objective of our study is to compare the feature lists chosen by experts and ML algorithms, and point at the merits of this type of human-machine collaboration, we will primarily focus on the other three models and discuss the full model's performance hereafter.

As a robustness check, we performed the whole prediction process using two additional classifiers (AdaBoost and Neural Network classifiers), resulting in similar gaps between the feature sets, yet yielding inferior predictive power (Online Appendix A). These results further strengthen our findings.

Examining the three confusion matrices, the ML-model provided more FP cases than the expert and the collaborative models (111 vs. 49 and 69, respectively), yet when combined with the expert list the number of FP is dramatically reduced. Also evident in the matrices is the relatively small number of TP cases identified by the expert-list model, 12, compared with 34 and 30 in the ML and collaborative models, respectively. The expert model, however, identified more TN cases than the other two models.

## 4. Discussion

The current study compared the performance of a prediction algorithm in predicting 30-day readmission of CHF patients for the same diagnosis, using three feature lists: experts list, ML list and a merged list. Although the ML list contained only 21 features as opposed to 26

**Table 5**
Comparison of the predictive power of the three feature lists for 30-day readmission.

| Model | Accuracy | F1 score | Recall | Precision | AUC [C.I. 95%] | TP | FP | TN | FN |
|---|---|---|---|---|---|---|---|---|---|
| All 120 variables | **0.8889** | 0.1672 | 0.1182 | 0.2857 | **0.8378** [0.818–0.857] | 24 | 60 | 1890 | 169 |
| Expert List | 0.8885 | 0.0909 | 0.0591 | 0.1967 | 0.7116 [0.679–0.744] | 12 | 49 | 1901 | 191 |
| ML List | 0.8699 | 0.1954 | **0.1674** | 0.2344 | 0.8156 [0.794–0.837] | 34 | 111 | 1839 | 169 |
| Collaborative list | 0.8876 | **0.1987** | 0.1478 | **0.3030** | 0.8289 [0.807–0.851] | 30 | 69 | 1881 | 173 |

Notes: C.I. 95% - Confidence Interval 95%; TP-True Positive; FP- False Positive; TN- True Negative; FN- False Negative; The best AUC result is bolded.

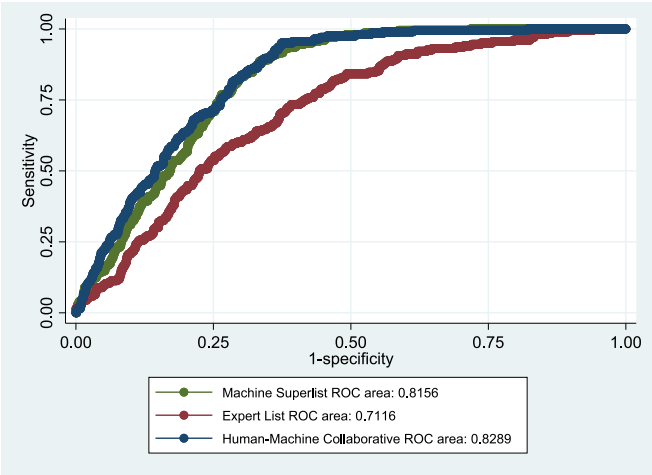


**Fig. 2.** Comparison of the AUCs for the prediction of 30-day readmission.

features in the expert list, it performed significantly better in terms of AUC. The merged list, however, with 42 features, performed slightly better than the ML list, implying, pending further research, the merit of this type of collaboration. Noticeably, the model based on the merged list significantly reduced the number of FP cases compared to the ML-based model, while performing similarly in FN rates. This is important when prediction models are implemented in practical tools such as CDSSs that are intended to augment experts' capability to precisely identify high-risk patients.

We have also evaluated the model with all 120 features to ensure completeness, and the results show slightly improved performance in terms of AUC and accuracy. Although improving the performance could be advantageous, particularly for studies that aim to develop an optimal prediction model, our study focuses on identifying and evaluating the performance obtained by a key set of features that are selected by domain experts and ML methods. Therefore, we suggest further research to explore the potential benefits of incorporating additional features in developing more robust and accurate prediction models.

Interestingly, while the experts chose features from five different domains (personal, medication, comorbidities, ECHO and lab results), the ML algorithms produced a more focused set of features based only on ECHO and lab results. The diverse expert list is well aligned with feature lists used in other models predicting 30-day readmission, as demonstrated in Table 1. For example, the LACE, a commonly used index to predict risk of unplanned readmission within 30 days after hospital discharge, is composed of the length of stay in the current hospitalization, 14 comorbidities, number of previous hospitalizations, and number of emergency department visits [127]. Similarly, a study examining features affecting readmission of patients aged 80 or older elicited the following features: any degree of impaired cognition, nursing home residence, CHF, and creatinine level > 1.5 mg/dl [23]. A meta-analysis pooling 144 features from 69 studies revealed that ten features were significantly associated with readmission within 90 days post discharge (chronic lung disease, chronic kidney disease, atherosclerotic vascular disease (peripheral, coronary, and cerebrovascular), diabetes, anaemia, lower systolic blood pressure, previous admission, multidisciplinary treatment, and use of beta-blockade and angiotensin-converting enzyme inhibition or angiotensin receptor blockade), mostly comorbidities and medication [103]. A close examination of the literature reveals that the feature sets affecting CHF patients' readmission were rather parsimonious and tended to include mainly comorbidities. This study thus proposes a somewhat different approach, where a relatively limited set of lab and imaging (ECHO) results (21 features), selected by ML algorithms, performed better in terms of prediction of 30-day readmission than a more diverse list selected by experts that included comorbidities as well as lab results. Lab results and imaging are not only easily available nowadays in most hospitals, but they are also quite often automatically populated into the EHR, making it less error prone. These results are also accumulated along the temporal axis of the patient's hospitalization history, allowing analyses that consider the time dimension, as well, and are likely to more sensitively reflect the health status of each subject in the studied cohort. Thus, since most studies in the domain of readmission prediction rely largely on comorbidities, medications, and a small number of common markers [39,41,75,80,115] as demonstrated in Table 1, and only a handful of studies attest to the improved performance of prediction models based largely on lab results [4,59], our study contributes to this relatively scarce line of research that leverages routinely collected and automatically updated objective lab data on disease status and disease progression.

Although prior research investigated the topic of human-machine collaboration, e.g. [88,139] (see also section 1.3), we found only one study that examined this collaboration in the domain of feature selection for risk prediction models [9], albeit differently. Ashfaq, et al. [9] detail only the human-selected features and then show that adding some of these features to a neural network prediction model improves its performance, without detailing the difference between the two lists. In contrast, the current study demonstrates that the human and ML lists have only three features in common, and examines the performance of these lists in prediction models.

Although a set of 42 features seems quite voluminous and perhaps less practical for routine use by clinicians, it might be easily fed into ML algorithms to provide more accurate predictive power in clinical decision making. Prediction models have served as the foundation and building blocks of modern clinical DSSs (CDSSs) [5], and we envision our and similar models to pave the way to future CDSSs, where a limited set of indicative features from the patients' EHR will be automatically fed into a CDSS to assist and augment physicians' capabilities to identify high-risk patients at the point-of-care [15]. Such an application raises the issue of the number of features that should be included. Theoretically, ML-based prediction models can use the full feature-set without a preliminary feature selection stage. In our case, a valid question is why not use the full list of 120 variables? An examination of a model with the full 120 feature-set (see Table 5) showed that the gain in AUC compared to the collaborative list of 42 features is minimal. Furthermore, based on the literature [16], and on views of physicians we have talked to, a large feature set overwhelms physicians, rendering a sense of information overload and distrust in a system with unexplainable results [3,65,95], likely resulting in suboptimal decision making [113]. Moreover, information overload may cause decision makers to not distinguish between essential and insignificant information [95,125], as well as utilize less information [78,123] while using different search patterns [117].

Overall, information overload may well reduce the effectiveness of decision makers [93], therefore finding the optimal feature set for a purposed target is important [106,118]. Another consideration when contemplating the full feature list against a reduced list is the relatively strong computing power required when running omni-dimensional datasets, that also take a long processing time. More importantly, models with a large number of features do not provide explainable results (see section 1.4), therefore are unlikely to be accepted by physicians as part of their routine practice [116]. To overcome this limitation, explainable AI tools and algorithms could be harnessed in future research. Hence, we conclude that although not essential in studies aimed at developing best prediction models, a pre-processing stage of feature selection can be beneficial when the prediction model is developed with practical or theoretical intentions. In choosing between the ML-list and the merged list we argue that although the merged list's contribution to the model's AUC is quite small compared to the ML-based model, it carries, in our case, two merits: 1) it expands the original set of features from just one category (lab results) to other categories, such as comorbidities and medications that are more familiar to practitioners, and 2) at least in our case, it lowers the number of FP, an unwarranted error in a precision-care environment.

When comparing the performance of the three models to other models appearing in the literature, we conclude that while the expert-list model performance in terms of AUC (0.71) is similar to the average reported AUC (see Table 1), the other two models outperformed most other models predicting 30-day readmission (0.82 and 0.83, respectively). The widely used LACE model (LACE≥10 is considered high-risk) [31] can serve as a performance reference level. LACE's AUC ranges between 0.63 and 0.78 [7,11,14,109]. Likewise, a recent literature survey [17] showed that the AUC results of models predicting 30-day readmission of CHF patients ranged from 0.6 to 0.8, with only two models showing AUC levels above 0.8. The CAPE model that relies on routine EHR data as input to create a "Clinical Analytics Prediction Engine (CAPE)" [108] achieved an AUC levels of 0.69–0.71 in predicting readmission, considered a moderate discriminative level. This model can also serve as a reference to our work, due to the similarities in the data source and target variable. In sum, our ML and collaborative models outperform most models published in the extant literature, while the expert-list model provides the average performance."

### 4.1. Contribution

The present study addresses four key healthcare information systems challenges: 1) the call for process innovation in healthcare, particularly by harnessing the plethora of health data that is currently available to improve outcomes, effectiveness and efficiency, 2) harnessing ML techniques and algorithms within big-data analytics to improve healthcare decision making, 3) addressing the issue of the bounded capacity of humans to effectively and efficiently consume large amount of data, and 4) an alternative human-machine collaboration concept, where two entities do not necessarily perform the task simultaneously but, instead, initially utilize the different types of expertise independently to subsequently produce a better merged outcome.

Addressing the call for process innovation [2,13,26,63,66], we envision clinicians relying more heavily on patient-level data in clinical decision making, alongside their expertise. As evident by the better performance of the shorter, mostly lab-related, feature set selected by ML, it is perhaps timely to bring new capabilities into current medical processes. Such capabilities are, for example, harnessing ML alongside human expertise, addressing the second challenge. Many data analytics examples exist showing improved performance in outcomes prediction, by harnessing ML techniques and algorithms within big-data analytics [10,30,76,96,129]. Examples are in kidney transplants using Bayesian decision support model [120], monitoring Alzheimer patients' medical conditions using intelligent agent [30], predicting heart transplant survival probabilities using isotonic regression [34] and disease management for lower back pain via autoregressive moving-average [87]. Moreover, creating bias-aware algorithms can compensate for biases created by human-generated data [3], thus our results demonstrate an alternative method – combining the products of the humans and machine to create a more balanced and comprehensive product. We suggest that the merged list that contains features selected by experts taken from domains other than lab results, might be more readily accepted by clinicians, as well as serve the identification of risk features while not impeding the performance of the prediction algorithm. This process innovation, introducing an alternative type of human and machine collaboration, aims at addressing the cognitive burden of data overload, caused by the bounded capacity of humans to effectively and efficiently consume large amount of data [113], most relevant in managing chronic diseases [84]. Big data analytics can compensate for this human deficiency by including more variables in the prediction models, thereby achieving higher predictive power that can improve clinical decision making. These models are not only more accurate for patients' risk stratification, but also mitigate biases associated with human cognition [3], while taking into account multiple criteria [56]. As noted above, demonstrating the merits of a different concept of human-machine collaboration is perhaps the most innovative contribution of this study. Whereas this concept generally relates to humans and machines performing a task together, we show that working in parallel and then combining the results might be beneficial, when the results are different. Thus, the process utilizes the expertise of the two entities for better results, allowing humans to examine the machine results, reflect, and perhaps modify their results, in an ongoing process until a parsimonious set is achieved. This could be translated into an actionable tool that might impact experts' knowledge in the Cardiology environment and beyond.

The study can be analyzed from two of the three perspectives of human-machine collaboration as pointed out by Xiong, et al. [132]: the organizational and relational perspectives. The organizational perspective deals with task allocation between humans and machines, and early research advocated optimal allocation by capabilities: assign to each party what they do best [40]. More recent studies argued that humans and machines should no longer be analyzed separately but as a hybrid team, taking into account psychological issues related to how humans perceive teaming up with machines to perform tasks that traditionally required their expertise [131]. The collaboration proposed in the study combines the expertise of both humans and machines hence can be regarded as a teamwork, creating a superior feature list for readmission prediction. This perspective aligns well with the concerns of clinicians who are reluctant to use AI-driven CDSSs due to anxiety over losing the impact of their expertise [116]. The relational perspective focuses on the hierarchy between the humans and machine, generally placing the humans as principal decision-makers, assisted by machines [54]. Recent research advocated human-centered AI [133], where machines are designed to augment human intelligence rather than replace it [57,94]. We suggest that the proposed human-machine collaboration can augment experts' intelligence due to the substantial difference between the two feature lists. By examining the ML list, experts can reflect on the optimal feature list, thereby consider assessing patients' readmission risk based on a temporal, longitudinal, examination of lab results, rather than on local clinical measures and comorbidities. A CDSS based on such human-machine collaboration can contribute to more positive attitude of clinicians towards AI-driven CDSSs, designed under the humans-in-the-center concept. The third perspective of human-machine interaction is closely related to the interface through which humans interact with AI-driven solutions. This perspective is beyond the scope of this study but can be explored in future research.

## 5. Conclusions and suggestions for future research

The objective of the current study was to explore the feasibility of linking ML based predictive analytics with unique knowledge of human

experts. The results contribute to research in several ways. First, we demonstrate the use of four feature selection algorithms that are based on ML and big data analytics. These are useful for analysis of health databases that are continuously becoming larger, multi-dimensional and rather complex. Second, the results show that different algorithms select different features, hence researchers should acknowledge the importance of the analytic tools. Third, we use XGBoost as the prediction algorithm, a tool less frequently used for medical analysis. Fourth, we achieved AUC scores ~80% for the ML and collaborative lists, higher than findings in previous studies, where most AUCs were ~ 70%. It is proposed that future research further examines each one of these contributions on different datasets.

For practice, the results contribute to clinicians' acknowledgement of the strength of lab and imaging results in patients' risk stratification. The prevalence of these data as part of the EHR eases their use in health data analytics, and the results of this study demonstrate their adequacy as substitution to comorbidities data. These are likewise longitudinal data, which facilitate temporal analysis rather than just a snapshot in time. With time and improvement of health information technology these data will not only be prevalent, but also more accurate. Hence future research should look more closely into lab and imaging data as predictors pf patients' risk status. Better risk stratification is likely to improve clinical decision making at the point of care and improve personalized medicine.

Another important avenue for future research of a more clinical nature is the transformation of feature sets into risk features. While a feature set is an input to a prediction algorithm, it nonetheless indicates which features most significantly affect the adverse event targeted. Mapping the feature set into various combinations of feature values might provide a risk-factor matrix that clinicians could use for risk stratification during routine clinical decision making.

An important observation of this study is the significant difference between the experts and the ML feature sets. We conjecture that the differences stem from the fact that comorbidities as features in a prediction model, are static, categorical (Y/N) variables. In contrast, lab results that are proxies of comorbidities, are dynamic and time-varying, thereby might perform better in prediction models. Nonetheless when selecting features that affect readmission, clinicians can process data on comorbidities and personal traits more easily than lab results, since the latter are stored in a format that is often difficult for humans to process. Interviewing experts to elicit their interpretation of the ML list and understand their perceptions of the differences between the two lists is an important topic for future research. Another important avenue for future research is the design of CDSSs that embrace the human-centered AI-driven strategy in a manner that will be readily adopted by clinicians. Finally, given the impressive performance of ML models with larger feature sets, it is recommended that future research explore the optimal feature set without constraining the number of features. Explainable AI tools and algorithms can then be employed to identify the most important features for clinicians to consider in their clinical decision-making.

Although our paper shows that the model with the complete feature set (comprising of 120 variables) only marginally outperformed the other three models, it is essential to conduct additional research on different feature sets to validate these findings and avoid potential overfitting.

Several limitations of this study should be noted. First, the significant differences in feature ranking among the four FS algorithms is intriguing. For example, 'calculated osmolality' is ranked 1 by the two logistic regression algorithms, whereas the random forest ones rank it 43 (Gini) and 19 (IG). Other similar cases are observed as well (see Table 3). Combining these rankings for better feature selection using advanced statistical approaches presents another opportunity for future research.

Second, while the framework and tools are generalizable, we call for caution in generalizing the results, since our study focuses on one institution (despite the size and richness of the dataset). Third, over the data period of eight years there might have been some changes in treatment methods and medications that were not accounted for in this study. The literature indicates a change in the early 1980s and 1990s [73], yet points at stagnation during the following decade [32]. Likewise, a senior cardiologist at Sheba who was consulted on this issue was not aware of major changes that could have affected the results.

Nonetheless, we believe that the methods and concepts employed here can be replicated with similar datasets from other medical domains, such as diabetes, renal diseases, organ transplantation, etc. using target variables such as mortality, transplant rejection, and others. Furthermore, the topic of human-machine collaboration is intensely researched in domains outside medicine. For example, Illankoon and Tretten [55] investigated artificial intelligence (AI) and human experts collaboration in the maintenance domain, and Kumar and Sharma [70] proposed a framework to improve human-machine collaboration in general, demonstrating the universal nature of the human-machine collaboration challenges. Dror, et al. [36]'s study of human-machine collaboration is rooted in the forensic decision making domain, and Ärje, et al. [8] compared the performance of human experts with ML in Taxonomic recognition. Thus, while the content might be different, concepts and challenges associated with human-machine collaboration are generalizable with degree of caution. Specifically, and related to FS by humans and ML, we suggest that it is possible to derive an expert-selected feature set for each domain and target variable, and then compare it to a ML-derived set, as in this study. We also note the absence of cardiologists' perceptions about the differences between the ML and expert lists. Future research needs to replicate our methods in other settings and domains to strengthen this study's external power, as well as include interviews with cardiologists. Similarly, the results should be replicated on different datasets to rule out the risk of overfitting, particularly with the larger, collaborative feature set. Finally, we suggest that future research examines the sensitivity of the feature selection process to the type of tools used in the analysis.

## CRediT authorship contribution statement

**Ofir Ben-Assuli:** Conceptualization, Methodology, Software, Validation, Formal analysis, Resources, Writing – original draft, Writing – review & editing, Visualization, Project administration. **Tsipi Heart:** Conceptualization, Methodology, Software, Validation, Investigation, Resources, Writing – original draft, Writing – review & editing, Visualization. **Robert Klempfner:** Conceptualization, Validation, Investigation, Resources, Data curation, Writing – original draft, Writing – review & editing, Supervision. **Rema Padman:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Supervision.

## Declaration of Competing Interest

None, for all authors.

## Data availability

The data that has been used is confidential.

## Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.dss.2023.113982.

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